

AI/ML Interview Preparation Guide

House Of Couton — SME AI/ML Internship | Advanced Technical Assessment

CORE ML CONCEPTS

Q1. What is the difference between supervised, unsupervised, and reinforcement learning?

- Supervised Learning: Uses labeled data. Model learns input-output mapping. Ex: Email spam detection (labeled spam/not spam).
- Unsupervised Learning: Uses unlabeled data. Finds hidden patterns. Ex: Customer segmentation by purchasing behavior.
- Reinforcement Learning: Agent learns via rewards/penalties interacting with environment. Ex: Self-driving cars, game-playing AIs (AlphaGo).

★ Tip: Mention 'policy' for RL — the agent learns an optimal policy through trial and error.

Q2. What is overfitting? How do you detect and prevent it?

- Definition: Model memorizes training data (including noise) instead of learning general patterns.
- Detection: High training accuracy but significantly lower validation/test accuracy.
- Prevention: Collect more data, cross-validation, L1/L2 regularization, reduce model complexity, early stopping, dropout, data augmentation.

★ Tip: Explain Dropout specifically — randomly deactivates neurons, forcing the network to learn redundant representations.

Q3. Explain Precision, Recall, and F1-Score. When would you prioritize Recall?

- Precision = $TP / (TP + FP)$ — Of all predicted positives, how many are truly positive?
- Recall = $TP / (TP + FN)$ — Of all actual positives, how many did we catch?
- F1-Score = $2 \times (Precision \times Recall) / (Precision + Recall)$ — harmonic mean of both.
- Prioritize Recall when false negatives are dangerous: cancer detection, fraud detection.
- Prioritize Precision when false positives are costly: spam filtering, recommendation systems.

★ Tip: Always link to a concrete medical or financial example — interviewers love this.

Q4. What is the bias-variance tradeoff?

- Bias: Error from wrong assumptions. High bias = underfitting (model too simple).
- Variance: Error from sensitivity to data fluctuations. High variance = overfitting (model too complex).
- Tradeoff: Decreasing bias tends to increase variance and vice versa.
- Goal: Find the sweet spot with low bias AND low variance — achieved via regularization, ensemble methods, or proper model selection.

★ Tip: Draw the U-curve mentally — total error = $bias^2 + variance + irreducible\ noise$.

DEEP LEARNING & TRANSFORMERS

Q5. Explain the Transformer architecture. What problem did it solve vs RNNs?

- RNN problems: Sequential processing (slow), difficulty capturing long-range dependencies, vanishing gradients.
- Transformer solution: Replaces recurrence with self-attention. Processes all tokens in parallel — much faster and more scalable.
- Attention mechanism: Each word attends to all other words, weighted by relevance.
- Multi-head attention: Multiple attention heads capture different types of relationships simultaneously.
- Positional encoding: Since no recurrence, the order of tokens is encoded explicitly.
- Foundation for GPT, BERT, T5, LLaMA, and all modern LLMs.

★ *Tip: Mention 'positional encoding' and 'multi-head attention' to show technical depth.*

Q6. What is the difference between CNN, RNN, and LSTM?

- CNN (Convolutional Neural Network): Best for spatial data (images). Uses convolution filters to detect local patterns. Efficient due to weight sharing.
- RNN (Recurrent Neural Network): Designed for sequential data. Has memory via hidden state passed between steps. Suffers from vanishing gradients.
- LSTM (Long Short-Term Memory): Improved RNN with gates (input, forget, output) to selectively remember/forget information. Handles long-range dependencies better.

★ *Tip: Add GRU as a lighter alternative to LSTM with fewer parameters.*

NLP & MODERN AI

Q7. What is RAG (Retrieval-Augmented Generation)? How does it work?

- Problem it solves: LLMs have a knowledge cutoff and can hallucinate. RAG grounds responses in real, up-to-date documents.
- How it works: 1) User query is embedded as a vector. 2) Similar documents are retrieved from a vector store (ChromaDB, FAISS, Pinecone). 3) Retrieved context + query are passed to the LLM. 4) LLM generates a grounded, factual response.
- Key components: Embedding model (e.g. all-MiniLM-L6-v2), Vector DB, LLM (e.g. GPT-4, LLaMA).
- Example: Second Brain — knowledge management system using ChromaDB + LangChain + Groq LLaMA-3.

★ *Tip: Mention hybrid search (vector + keyword) and re-ranking for advanced RAG pipelines.*

Q8. What are word embeddings? What is the difference between Word2Vec and BERT embeddings?

- Word embeddings: Dense vector representations of words capturing semantic meaning. Similar words have similar vectors.
- Word2Vec: Static embeddings — each word has ONE fixed vector regardless of context. 'Bank' always has the same vector.

- BERT embeddings: Contextual embeddings — the vector for 'bank' changes depending on the sentence (river bank vs. financial bank).
 - Modern use: Sentence Transformers (e.g. all-MiniLM-L6-v2) for semantic similarity and RAG pipelines.
- ★ *Tip: Mention that BERT uses bidirectional attention while GPT uses unidirectional (causal) attention.*

Q9. What is the difference between LangChain and LangGraph?

- LangChain: Framework for building LLM pipelines. Good for linear chains of operations (prompt -> LLM -> output).
- LangGraph: Extension for stateful, multi-agent workflows. Models pipelines as directed graphs with nodes and edges.
- When to use LangGraph: Complex workflows with loops, conditional branching, or multiple collaborating agents.
- Real result: In Second Brain project, LangGraph achieved correctness score of 0.79 vs LangChain's 0.60 on 31 QA pairs.

★ *Tip: This is a strong differentiator — few candidates have hands-on comparison experience like this.*

Q10. How would you evaluate an LLM-based application? What metrics would you use?

- Automated metrics: BLEU, ROUGE (text similarity), BERTScore (semantic similarity), Exact Match.
- LLM-as-judge: Use a stronger model (GPT-4) to score responses for correctness, relevance, faithfulness.
- RAG-specific metrics: Context Precision, Context Recall, Answer Faithfulness, Answer Relevancy (RAGAS framework).
- Human evaluation: Gold standard for nuanced quality, especially for open-ended tasks.
- Tooling: LangSmith for tracing, monitoring, and evaluation of LangChain/LangGraph pipelines.

★ *Tip: Mention LangSmith specifically — shows familiarity with production-grade LLM observability tools.*

BONUS: LIKELY FOLLOW-UP QUESTIONS

Bonus Q1. What is gradient descent? What are its variants?

- Gradient Descent: Optimization algorithm that minimizes loss by moving in the direction of the steepest decrease (negative gradient).
- Batch GD: Uses entire dataset per step. Stable but slow.
- Stochastic GD (SGD): Uses one sample per step. Fast but noisy.
- Mini-batch GD: Uses small batches. Best of both worlds. Most commonly used in practice.
- Adaptive optimizers: Adam (adaptive learning rates), RMSprop — faster convergence in practice.

★ *Tip: Adam = Momentum + RMSprop. It's the default optimizer for most deep learning tasks.*

Bonus Q2. What is transfer learning? Give an example.

- Definition: Using a pre-trained model (trained on large data) and fine-tuning it on a smaller, task-specific dataset.
- Why it works: Early layers learn general features (edges, shapes, syntax) reusable across tasks.

- Example (NLP): Fine-tuning BERT on a sentiment analysis dataset instead of training from scratch.
- Example (CV): Fine-tuning ResNet on medical images for disease classification.

★ *Tip: Mention 'frozen layers' — often we freeze early layers and only train the top layers during fine-tuning.*